Conservation Genetics (2024) 25:291–304
https://doi.org/10.1007/s10592-023-01572-9

RESEARCH ARTICLE

Conservation genomics of the threatened Trispot Darter (*Etheostoma trisella*)

Kayla M. Fast[1,2] · Brook L. Fluker[3] · Bernard R. Kuhajda[4] · Patrick E. O'Neil[5] · Stuart W. McGregor[5] · Matthew S. Piteo[6] · Michael W. Sandel[2,7]

Received: 9 September 2022 / Accepted: 5 September 2023 / Published online: 26 September 2023
© The Author(s), under exclusive licence to Springer Nature B.V. 2023

Abstract

The southeastern United States is a hotspot of biodiversity, but aquatic habitats are fragmented by anthropogenic activities such as hydrologic alteration. Small-stream-inhabiting fish can suffer population declines, loss of genetic diversity, and migration impediment from riverine impoundment. The Trispot Darter (*Etheostoma trisella*) is a small, freshwater fish endemic to the southeastern United States. *E. trisella* was previously believed to be extinct and has now been listed as threatened under the U.S. Endangered Species Act, due to destruction of its limited habitat and a restricted range. We used mitochondrial DNA (NADH dehydrogenase subunit 2 gene), seven microsatellite loci, and 9732 single nucleotide polymorphisms (SNPs) to evaluate population structure and diversity in *E. trisella.* Mitochondrial data provided evidence of historical connectivity between populations, with haplotype sharing across populations and weak support for population structure. Microsatellite and SNP data, however, indicate that populations have more recently become isolated. Furthermore, we detect three distinct management units (i.e., genetic groups) which reflect isolated geographic localities (i.e., Little Canoe Creek, Ballplay Creek, and a system including the Conasauga and Coosawattee rivers). We also detected a recent bottleneck event in the Ballplay Creek population as well as a low effective population size. Tests of isolation by distance further suggest that populations are structured by riverine isolation rather than geographic distance. A better understanding about the distribution, abundance, and habitat occupancy of all *E. trisella* populations will be important for informing future decisions for conservation of the species.

Keywords Conservation genetics · Microsatellites · mtDNA · SNPs

✉ Kayla M. Fast
kmf160@msstate.edu

✉ Michael W. Sandel
mws297@msstate.edu

1 Department of Biological and Environmental Sciences, The University of West Alabama, Livingston, AL, USA

2 Department of Wildlife, Fisheries and Aquaculture, Mississippi State University, Mississippi State, MS, USA

3 Department of Biological Sciences, Arkansas State University, Jonesboro, AR, USA

4 Tennessee Aquarium Conservation Institute, Chattanooga, TN, USA

5 Geological Survey of Alabama, Tuscaloosa, AL, USA

6 U.S. Fish and Wildlife Service, Abernathy Fish Technology Center, Longview, WA, USA

7 Forest and Wildlife Research Center, Mississippi State University, Mississippi State, MS, USA

Introduction

Etheostoma trisella, the Trispot Darter, is a small, freshwater fish endemic to the Coosa River system (Fig. 1) in the southeastern United States including northeastern Alabama, northwestern Georgia, and southeastern Tennessee (Page and Burr 2011). *E. trisella* was described based on a single specimen collected in 1947 at Cowans Creek in Alabama (Bailey and Richards 1963). In 1960, this locality was inundated by Weiss Lake, and *E. trisella* was considered extinct. Later a second museum specimen of *E. trisella* was discovered from the Coosa River in Alabama which was collected in 1958; this locality was inundated by construction of H. Neely Henry Lake in 1965. In 2004, *E. trisella* was presumed extirpated in Alabama after several surveys did not yield any specimens (Mettee et al. 1996; Neely and Mayden 1999; Warren 2004; Boschung and Mayden 2004). *E. trisella* was recently rediscovered in Alabama in

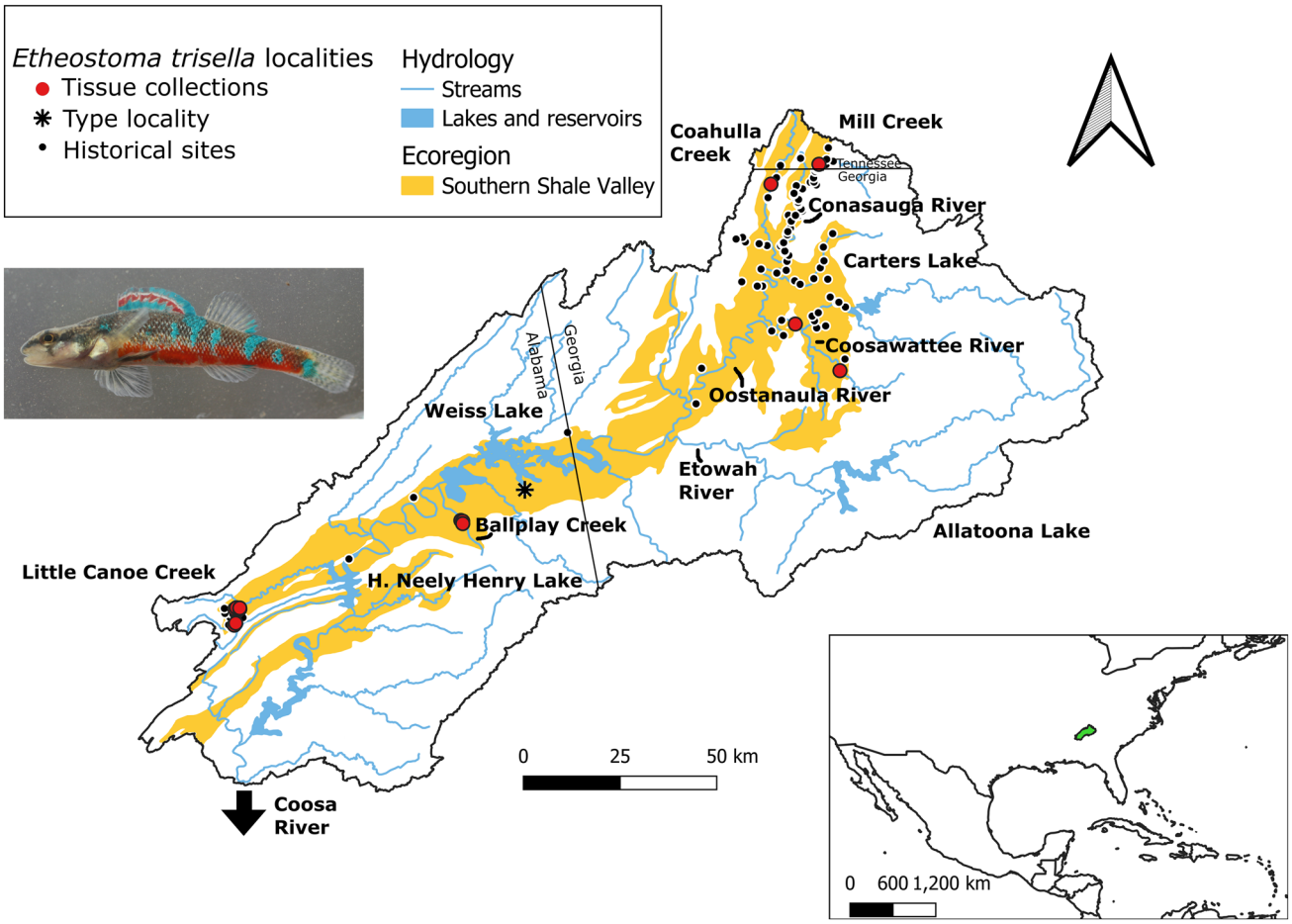

Fig. 1 Collection localities within the Upper Coosa River watershed for *Etheostoma trisella* used in mitochondrial DNA, microsatellite, and SNP analyses. The type locality at Cowans Creek is indicated with an asterisk and an arrow is used to show where the watershed drains. Inset photograph: male *E. trisella* photographed at Ballplay Creek by Bernard R. Kuhajda. The colored polygon in the inset map depicts the Upper Coosa River watershed

the Little Canoe Creek and Ballplay Creek systems (O'Neil et al. 2009; Johnson et al. 2013; Bearden et al. 2021). A museum specimen of *E. trisella* collected in 1954 was discovered from a Conasauga River tributary in Georgia, with subsequent collection of specimens at this site and at another site in the Conasauga River system in the 1960s (Etnier and Starnes 1993). The currently recognized range in Georgia and Tennessee includes the Conasauga River system upstream of the confluence with the Coosawattee River, including Mill, Coahulla, and Holly creeks, in the Coosawattee River proper below Carters Lake, and the Oostanaula River between Arumuchee Creek and Calhoun (Boschung and Mayden 2004; Bearden et al. 2021; Georgia Department of Natural Resources 2023). Surveys between 2007 and 2017 recovered *E. trisella* in only 20% of its historical range (U.S. Fish and Wildlife Service 2017).

E. trisella is one of only a few darters that undergo migration between two seasonally connected habitats (Williams and Robison 1980; Ryon 1986). During the non-breeding season (mid-April to mid-October), *E. trisella* disperses and occupies zones peripheral to the main channel of small- to medium-sized streams, characterized by the presence of gravel, vegetation, and a slow-moving current. Seasonal rainfall allows the darter to collectively migrate upstream to breeding grounds where they spawn from January to March. This habitat consists of small, groundwater-fed ephemeral streams or seepage areas with groundwater input in pastures and floodplains containing high amounts of vegetation, root masses, or detritus. Because *E. trisella* is a migratory fish, gene flow between populations can be significantly impacted if movement between seasonal habitats is impeded.

Given that *E. trisella* has a severely limited distribution and is highly impacted by anthropogenic activity, the darter was listed as Threatened under the US Endangered Species Act (ESA; Rule 83 FR 67131) in 2018. The US states Alabama, Georgia, and Tennessee individually list *E. trisella* as a species of concern, threatened, and endangered, respectively (Freeman and Hagler 2016; Withers

2016; Shelton-Nix 2017). The American Fisheries Society's Endangered Species Committee also designates *E. trisella* as endangered (Jelks et al. 2008). The U.S. Fish and Wildlife Service (USFWS) currently recognizes four Management Units (MUs) for *E. trisella*: (1) Little Canoe Creek; (2) Ballplay Creek; (3) Conasauga River; and (4) Coosawattee River (USFWS 2018a, b). There are many differing criteria for conservation units (Fraser and Bernatchez 2001); here, an MU is one or more watersheds currently occupied by *E. trisella* with similar management strategy requirements and genetic similarity within the watershed (USFWS 2017). The Species Status Assessment (SSA, USFWS 2017) for *E. trisella* noted that MUs 1–3 were established based on unpublished microsatellite DNA (μsDNA) data analyses (see Fluker and Kuhajda, unpublished data) that recovered each as a genetically distinct group. The unpublished dataset of Fluker and Kuhajda (unpublished data) did not contain samples from the Coosawattee River, and it is still unclear if this MU is genetically distinct from the other MUs. Despite uncertainty of genetic distinctiveness, the USFWS (2017) designated the Coosawattee River a unique MU due to the presence of a large hydroelectric dam (Carters Dam) that is located upstream of designated critical habitat for *E. trisella* (USFWS 2017, 2020). In addition, the H. Neely Henry Dam and Weiss Lake Dam and their associated reservoirs on the mainstem of the Coosa River represent barriers to connectivity among populations (USFWS 2017). Accordingly, hydrologic alteration and reduced connectivity were listed among the risk factors influencing the viability of *E. trisella* (USFWS 2018a, b).

The impoundment of aquatic habitats by dams negatively influences wildlife, and biodiversity is projected to decline in response to human socio-economic scenarios (Rands et al. 2010; Pereira et al. 2010; Wu et al. 2019). Dam construction changes the flow of water upstream from lotic to lentic water, which causes a loss of spawning grounds and changes the fish assemblage (Wu et al. 2019). Downstream effects include altered hydrology and temperatures, including the elimination of natural seasonal flooding of floodplains (Wu et al. 2019), which may reduce or eliminate access for *E. trisella* to spawning grounds. This threat is prominent for critical habitat in the Coosawattee River proper, which is located directly downstream of Carters Dam. Additionally, populations that are limited in size and isolated from others are affected most heavily by habitat fragmentation (Fischer and Lindenmayer 2007; Haddad et al. 2015). These small, isolated populations are vulnerable to extinction risk and loss of genetic diversity (Willi et al. 2006; Frankham 2015; Coleman et al. 2018). Indicators of negative impacts often are not apparent immediately and occur gradually over generations; therefore, irreversible loss of biodiversity may occur before it is detected (Haddad et al. 2015; Wu et al. 2019).

It is understood that fishes with restricted distributions are vulnerable to extinction risk. A logical question is how to use genetics and genomics to guide conservation efforts. When species show genetic population structure, this differentiation is often used to define conservation units (e.g., MUs; Doadrio et al. 1996; Mandimbihasina et al. 2020; Corral-Lou et al. 2021; Kim and Roe 2021, Magris et al. 2021). These units can then be used to guide conservation efforts by advising revisions to current plans (Johansen et al. 2020) and suggesting goals for population management (Knaepkens et al. 2004). Fishes with restricted distributions may show evidence of genetic differentiation due to ancient isolation processes or more contemporary events, and it has been suggested that effective conservation plans and habitat management target multiple scales including populations assigned as MUs (Lopes-Lima et al. 2017; Alcaraz and Gholami 2020).

In this study, we use population genetic and genomic approaches to better understand genetic structure, genetic diversity, and demographic history for *E. trisella*. We construct a comprehensive dataset, including unpublished mitochondrial DNA (mtDNA) and μsDNA data from Fluker and Kuhajda (unpublished data) and a reduced-representation genomic single nucleotide polymorphism (SNP) dataset using DArTseq™ technology. Our use of mtDNA, μsDNA, and SNPs provides an opportunity to evaluate different aspects of genomic history (Bohling et al. 2019) for *E. trisella* that may better inform conservation and management strategies for this rare species. We evaluate genetic diversity of *E. trisella* MUs to understand genetic diversity across the range of each MU, and we additionally measure genetic diversity of creek systems to determine if any are drastically different within a given MU.

Materials and methods

Sample collection

Etheostoma trisella was collected across the known range in the southeastern US. Sampling was conducted by backpack electrofishing and seining between 2008–2019 (Fig. 1 and Table 1). All collections conducted during this study were permitted by the Alabama Department of Conservation and Natural Resources and approved by the University of Alabama Institutional Animal Care and Use Committee (protocol numbers: 07–293-1; 07–293-2; 07–293-3). Collections primarily took place in breeding habitat during November through February to correspond with the spawning season of *E. trisella*. Three *E. trisella* from the Coosawattee River in Georgia were collected during the non-spawning season. Upon capture, a portion of the pectoral fin was removed, stored in 95–100% ethanol at -20 °C, and the fish released

Table 1 Creek system, number (*N*), and catalog numbers for samples of *Etheostoma trisella* used in analyses of mitochondrial DNA (mtDNA), microsatellite DNA (μsDNA) and single nucleotide polymorphisms (SNPs)

Creek system	*N* (mtDNA)	*N* (μsDNA)	*N* (SNPs)	Catalog number
Little Canoe Creek	3	3	1	UAIC 15488.01
	2	8	3	UAIC 15489.01, 15491.02
	5	30	7	UAIC 15550.02
	–	2	–	UAIC 15553.01
	–	3	2	UAIC 15554.01
	–	–	2	KJ181211.03
	–	–	3	UAIC 15765.01
	–	–	8	UAIC 15766.01
	–	–	10	UAIC 15764.01
	–	–	4	UAIC 15489.02
	–	–	7	UAIC 15767.01
	–	–	7	UAIC 15768.01, 15769.01
	–	–	3	UAIC 15770.01, 15772.01
Ballplay Creek	10	25	10	UAIC 15709.02, 15710.02, 15715.01
	–	3	2	UAIC 15711.01
	–	3	2	UAIC 15712.01, 15713.01
Coahulla Creek	10	29	9	UAIC 15669.01
Mill Creek	10	25	10	UAIC 15671.01
Coosawattee River	–	–	2	BA18.050
	–	–	1	BA11.038
Total	40	131	93	

University of Alabama Ichthyological Collection (UAIC); KJ181211.03 is stored in the University of West Alabama Zoological Collection; BA18.050 and BA11.038 are stored at Georgia Department of Natural Resources

immediately except for a subsample of whole specimens fixed in 10% formalin and stored in ethanol.

DNA extraction

Genomic DNA was isolated from tissue using the DNeasy Blood and Tissue Kit following the manufacturer's instructions (QIAGEN). The quality of DNA was verified by electrophoresis using a 1.5% agarose gel stained with ethidium bromide. DNA quantity was determined using a NanoDrop 2000 Spectrophotometer (Thermo Fisher Scientific). Purified DNA was stored at 4 °C until further analysis.

Mitochondrial DNA sequencing and analyses

The complete mtDNA NADH dehydrogenase subunit 2 (*ND2*) gene was amplified for 40 individuals (Table 1) using primers (Etheostomatinae external forward and external reverse) and conditions described in Lang and Mayden (2007). Amplified products were purified with the QIAquick PCR Purification Kit (QIAGEN) and sequenced in both directions with the amplification primers using an ABI PRISM® 3100 Genetic Analyzer (Applied Biosystems). Raw contigs were visually inspected for sequencing error, manually assembled into consensus sequences, and aligned by eye in BioEdit (Hall 1999; Hall and Alzohairy 2011). Genetic differentiation and a test for isolation by distance (IBD) based on mtDNA were assessed among samples using analysis of molecular variance (AMOVA) and a Mantel test in Arlequin v.3.5 (Excoffier and Lischer 2010). Tests for IBD used the shortest distance between sample locations along the river network. Since estimated mtDNA rates of molecular evolution have been established for darters (see Near et al. 2011; Fluker et al. 2014a, b), we used BEAST v.1.10.4 (Suchard et al. 2018) with coalescent Bayesian Skyline Plot (BSP) analysis to estimate the root age for all *E. trisella* mtDNA haplotypes and to estimate change in effective population size through time. BEAST analyses were run for 50 million generations with the HKY substitution model, strict clock calibrated with the *ND2* rate of 9.29×10^{-3} substitutions/site/my (Fluker et al. 2014a, b), and coalescent Bayesian Skyline tree prior with 10 groups and piecewise-constant model. Initial runs were also conducted with an uncorrelated relaxed lognormal clock (UCLN) model to assess the best fit. Tracer v.1.7.2 (Rambaut et al. 2018) was used to assess parameter convergence and to construct BSPs. We used *ND2* sequences to construct a haplotype network with TCS v.1.2 (Clement et al. 2000) to estimate genealogical relationships among haplotypes.

Microsatellite DNA genotyping and analyses

A total of 131 specimens (Table 1) was genotyped for nine tetranucleotide microsatellite loci using primers designed for other darter species from the genus *Etheostoma* (Tonnis 2006—Eca11EPA, Eca48EPA, Eca49EPA; Switzer et al. 2008—EosC6, EosC112, EosD11; Gabel et al. 2008—Esc18, Esc26b, Esc132b). Methods for allele scoring, and assessment of linkage disequilibrium (LD) and conformity to Hardy–Weinberg equilibrium (HWE) are described in Fluker et al. (2019). Summary statistics, AMOVA, and IBD tests were conducted with μsDNA data in Arlequin v.3.5. The number of genetically differentiated clusters (K) was estimated using STRUCTURE v.2.3.4 (Pritchard et al. 2000) with a model allowing admixture of genotypes, correlated allele frequencies among populations, and no prior population information. These STRUCTURE data were used to assign the MUs used in subsequent analyses. Ten replicate runs of 500,000 iterations (100,000 burn-in) were conducted with K values ranging from one to six. Most likely K values were determined based on a combination of the mean estimated ln probability of the data [lnP(K)] and the ad hoc statistic ΔK (Evanno et al. 2005). Tests for declines in effective population size (N_e) were evaluated with the heterozygosity excess method (BOTTLENECK v.1.2.02; Piry et al. 1999) and the M-ratio method (M_P_Val.exe, Critical_M.exe; Garza and Williamson 2001) as described in Fluker et al. (2019). Genetic diversity tests were conducted across creek systems and STRUCTURE assigned MUs. Measures of allelic richness in μsDNA were calculated in HP-Rare (Kalinowski 2005) by standardizing all populations to the smallest sample size.

DArTseq™ genotyping and analyses

Genotyping by DArTseq™ analysis was performed at Diversity Arrays Technology Pty, Ltd. (DArT; Canberra, ACT, Australia) utilizing DArT complexity reduction methods and next generation sequencing (Kilian et al. 2012). To optimize for *Etheostoma*, several enzyme combinations were tested for complexity reduction and the PstI-SphI method was selected. Digestion/ligation reactions were performed as described by Kilian et al. (2012) with modifications to the adaptors (Elshire et al. 2011). PstI-SphI fragments were amplified by PCR using the following reaction conditions: initial denaturation 94 °C for 1 min, 30 cycles of 94 °C for 20 s, 58 °C for 30 s, and 72 °C for 45 s, with final elongation at 72 °C for 7 min. Amplified products were applied to c-Bot (Illumina, San Diego, USA) bridge PCR and sequenced on an Illumina HiSeq2500 for 77 cycles.

Sequences were processed using proprietary DArT analytical pipelines. Poor quality sequences were filtered out using the following parameters: barcode region (Min Phred pass score 30, Min pass percentage 75) and whole reads (Min Phred pass score 10, Min pass percentage 50). Marker calling was completed using approximately 1,250,000 sequences per sample. DArT PL's proprietary SNP and SilicoDArT algorithms (DArTsoft14) were used for calling the presence/absence of restriction fragments in representation. The quality of SNP calls was assured by high average read depth per locus with an average across all markers over 10 reads/locus. The main selection criteria for high quality/low error rate markers was scoring consistency from technical replicates. Ninety-four samples were submitted for DArTseq™ (Table 1) and 16 had technical replicates which were used to estimate reproducibility of reported markers (> 99%). One sample did not pass quality-control filtering and was removed from analyses.

Multiple analysis methods were used with SNP data to evaluate patterns of population structure. Loading and filtering of SNPs were conducted using the package "dartR" in R v.4.1.1 (Gruber et al. 2018; R Core Team 2021). Using the "dartR" package, we performed a Principal Coordinate Analysis (PCoA) with results plotted using the two most informative principal coordinates. To identify and assign individuals to MUs, SNP data were transformed into the fastSTRUCTURE format for further analysis. From these data, ancestry matrices were computed using the R package "LEA" for $K = 1–10$ (Frichot and François 2015). The best-fit matrix was chosen based on the returned cross entropy value. These data from the LEA package were used to assign the MUs used in subsequent SNP analyses. We reconstructed a Bayesian phylogeny as described above except for use of the *ND2* rate of molecular evolution. TreeAnnotator v.2.6.0 was used to compile a single tree annotated with posterior probabilities. The resulting tree was viewed and annotated in FigTree v.1.4.4.

We evaluated the possibility of IBD using SNP data and the "ade4" package using a Mantel test and 999 permutations (Dray and Dufour 2007). For use in IBD, riverine distances between creek systems of *E. trisella* were calculated in the "riverdist" package and pairwise population fixation index (F_{st}) values in "dartR" (Tyers 2022). The SNP data in a genlight object were converted to the genepop file type for use in NeEstimator v.2.1 (Do et al. 2013) to estimate N_e for each MU. Effective population sizes were not calculated for creek systems because sample sizes were too small for the software to perform. Within Arlequin v.3.5, we evaluated the amount of population genetic structure with an AMOVA with 10,000 permutations and allowing up to 5.0% missing data. Between populations, we calculated pairwise population F_{st}. These F_{st} values were plotted in a heat map with a dendrogram using the R package "gplots" (Warnes et al. 2016). Within populations, we calculated observed heterozygosity (H_o) and expected heterozygosity (H_e) using only polymorphic loci and allowing up to 5.0% missing data.

Genetic diversity tests were conducted across creek systems and MUs, as defined by LEA. Differences in statistics between data types were determined using two-tailed t-tests.

Results

mtDNA

A total of 13 unique haplotypes were identified among the 40 individuals sequenced for the mitochondrial *ND2* gene (Online Resource 1; Table S1). Mill Creek is the most unique with respect to mtDNA as this population has three unique haplotypes and only shares haplotypes with Coahulla Creek. The other populations have 1–2 unique haplotypes, but there is a general pattern of haplotype sharing across populations (Online Resource 1; Table S1, Fig. S1). This pattern matched the finding of weak, but significant, structuring among creek systems and MUs based on AMOVA of mtDNA data (Tables 2 and 3). There was no relationship between mtDNA divergence and geographic distance among populations (Fig. 3; $r^2 = 0.24$; $p = 0.29$).

Coalescent phylogeographic reconstructions using BEAST with *ND2* data failed to resolve any of the four populations as monophyletic (Online Resource 1; Fig. S2). The only clades with significant support (> 0.95 posterior probability) grouped haplotypes from differing creek systems together and matched with closely related haplotype groupings in the haplotype network (i.e., haplotypes 1 + 4, 5 + 2, 6 + 12 + 13; Online Resource 1; Fig. S1, Fig. S2). The haplotype network estimated haplotype 3 as ancestral, which is shared between Little Canoe and Ballplay creek populations, and there were no more than five mutational steps among any two haplotypes (Online Resource 1; Fig. S1). Based on the root age estimate of the BEAST analysis, all sampled *E. trisella ND2* haplotypes coalesce to a common ancestor 187,952 [95% highest posterior density (HPD): 73,926–329,301] years ago. Because none of the populations were monophyletic based on *ND2* data, we present results of only the BSP analysis for the species as a single group. The BSP analysis of *ND2* data suggests that N_e for the species has remained relatively constant over the past 70,000 years (i.e., the mean N_e was relatively constant throughout time and the 95% confidence intervals were largely overlapping through time; Online Resource 1; Fig. S3).

Microsatellite DNA

Two of the nine µsDNA loci (Esc18 and EosD11) were monomorphic across samples and were excluded from analyses. The remaining seven loci were polymorphic, unlinked, and genotypes did not deviate from HWE. Evaluation of lnP(K) and ΔK distributions for STRUCTURE analyses were unambiguous for $K = 3$, suggesting three genetically

Table 2 Analysis of molecular variance (AMOVA) for different hierarchical groupings of *Etheostoma trisella* creek systems based on the mitochondrial (mt) *ND2* gene, seven microsatellite (µsDNA) loci, and 9732 single nucleotide polymorphisms (SNPs)

Data type	Source of variation	d.f.	SS	% of variance	Fixation index	*p* value
mtDNA	Among localities	3	16.78	28.55	$F_{ST} = 0.29$	< 0.00001
	Within localities	36	40.30	71.45	–	–
µsDNA	Among localities	3	46.93	8.74	$F_{ST} = 0.09$	< 0.00001
	Within localities	127	267.47	− 3.90	$F_{IS} = -0.04$	**0.99238**
	Within individuals	131	300.50	95.15	$F_{IT} = 0.05$	0.04406
SNPs	Among localities	4	30,514.40	11.50	$F_{ST} = 0.12$	< 0.00001
	Within localities	88	153,294.90	3.25	–	–
	Within individuals	93	150,534.80	85.25	–	–

[a]Degrees of freedom (d.f.); sum of squares (SS); non-significant results ($p > 0.05$) are shown in bold

Table 3 Analysis of molecular variance (AMOVA) for different hierarchical groupings of *Etheostoma trisella* management units (MUs) based on the mitochondrial (mt) *ND2* gene, seven microsatellite (µsDNA) loci, and 9732 single nucleotide polymorphisms (SNPs)

Data type	Source of variation	d.f	SS	% of variance	Fixation index	*p* value
mtDNA	Among localities	2	11.13	34.57	$F_{ST} = 0.22$	< 0.00001
	Within localities	37	49.95	78.23	–	–
µsDNA	Among localities	2	39.66	8.55	$F_{ST} = 0.09$	< 0.00001
	Within localities	128	274.74	− 3.04	$F_{IS} = -0.03$	**0.97495**
	Within individuals	131	300.50	94.49	$F_{IT} = 0.06$	0.04337
SNPs	Among localities	2	24,353.35	10.83	$F_{ST} = 0.08$	< 0.00001
	Within localities	90	159,007.66	4.07	–	–
	Within individuals	93	149,960.13	85.10	–	–

[a]Degrees of freedom (d.f.); sum of squares (SS); non-significant results ($p > 0.05$) are shown in bold

distinct groups based on our µsDNA data (Fig. 2). Individuals from Little Canoe and Ballplay creeks were each assigned to their own group, while Coahulla and Mill creeks were both assigned to a single group (i.e., Conasauga River; Fig. 2).

The number of alleles per locus ranged from 2 – 23 across all samples, with mean values lowest for Ballplay Creek (4.71), and highest for the Conasauga River MU (10.43). Estimates of genetic diversity for creek systems are listed in Table 4 and for MUs in Table 5. When compared across the four creek systems and three MUs, we found no significant differences in the mean number of alleles (*A*) or H_o. The AMOVA of microsatellite data revealed that 9% of the genetic diversity was distributed among creek systems (Table 2; $p < 0.01$) and 9% among MUs (Table 3; $p < 0.01$). The matrix of pairwise F_{st} estimates (Online Resource 1; Table S2 and S3) was generally consistent with STRUCTURE group assignments, and indicated low-moderate levels of genetic isolation among creek systems and MUs, all of which were statistically significant. There was no relationship

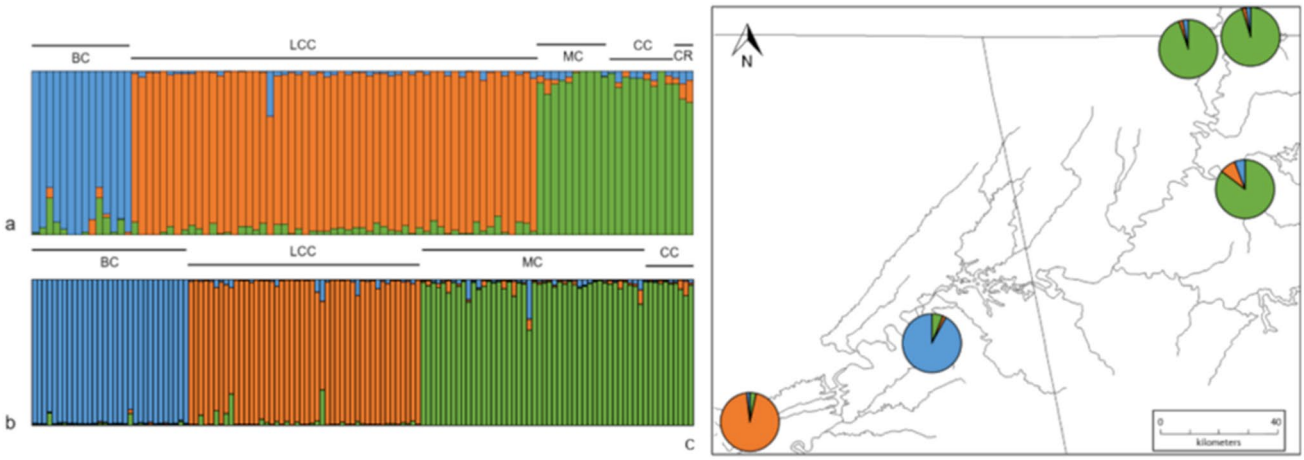

Fig. 2 Ancestry matrices estimated for *Etheostoma trisella* in: **a** LEA using 9732 single nucleotide polymorphisms (SNPs; $K = 3$) and **b** STRUCTURE using seven microsatellite loci ($K = 3$); **c** SNP ancestry estimates displayed as pie charts with geographic localities. *BC* Ballplay Creek, *LCC* Little Canoe Creek, *MC* Mill Creek, *CC* Coahulla Creek, *CR* Coosawattee River

Table 4 Genetic diversity estimates were calculated using seven microsatellite loci (µsDNA) and single nucleotide polymorphisms (SNPs) for *Etheostoma trisella* populations. Standard deviations (SD) are in parentheses

Creek system	H_o SNPs	H_o µsDNA	H_e SNPs	H_e µsDNA	*NA*	*A*	*AR*	*PA*	No. of loci
Little Canoe Creek	0.168 (0.174)	0.669 (0.31)	0.175 (0.175)	0.636 (0.30)	53	7.57	6.79	1.08	6997
Ballplay Creek	0.167 (0.204)	0.613 (0.25)	0.165 (0.188)	0.575 (0.33)	33	4.71	4.65	0.43	5112
Coahulla Creek	0.172 (0.202)	0.728 (0.15)	0.175 (0.187)	0.697 (0.35)	65	9.29	8.84	0.82	5493
Mill Creek	0.154 (0.184)	0.621 (0.30)	0.169 (0.183)	0.646 (0.34)	57	8.14	8.00	0.95	5496
Coosawattee River	0.159 (0.250)	–	0.170 (0.229)	–	–	–	–	–	3815

[a]Heterozygosity observed (H_o); heterozygosity expected (H_e); total number of alleles (*NA*); mean number of alleles per locus (*A*); allelic richness (*AR*); private allelic richness (*PA*); the number of SNP loci used to estimate heterozygosity (# of loci)

Table 5 Genetic diversity estimates were calculated using seven microsatellite loci (µsDNA) and single nucleotide polymorphisms (SNPs) for *Etheostoma trisella* management units (MUs). Standard deviations (SD) are in parentheses

MU	H_o SNPs	H_o µsDNA	H_e SNPs	H_e µsDNA	*NA*	*A*	*AR*	*PA*	# of loci
Little Canoe Creek	0.167 (0.174)	0.669 (0.31)	0.172 (0.175)	0.636 (0.30)	53	7.57	7.13	1.28	8636
Ballplay Creek	0.167 (0.205)	0.613 (0.36)	0.159 (0.188)	0.575 (0.33)	33	4.71	4.71	0.52	9251
Conasauga and Coosawattee Rivers	0.161 (0.164)	0.679 (0.36)	0.173 (0.167)	0.687 (0.35)	73	10.43	9.55	3.24	8335

[a]Heterozygosity observed (H_o); heterozygosity expected (H_e); total number of alleles (*NA*); mean number of alleles per locus (*A*); allelic richness (*AR*); private allelic richness (*PA*); the number of SNP loci used to estimate heterozygosity (# of loci)

between µsDNA divergence and geographic distance among populations (Fig. 3; $r^2 = 0.34$; $p = 0.13$). Results from the BOTTLENECK analysis revealed a significant excess of heterozygosity for Ballplay Creek ($p < 0.05$) suggesting the occurrence of a recent genetic bottleneck. Ballplay Creek also showed the lowest N_e (Table 6). No populations showed evidence for bottleneck based on *M*-ratio tests.

SNPs

After quality control and filtering of genotyping data by DArT, 93 of our 94 samples returned data usable for analyses. DArTseq™ data returned 20,673 SNP markers. We filtered out monomorphic loci (i.e., loci for which all individuals share the same genotype) that were introduced by the removal of one sample and obtained 20,669 SNPs (Gruber et al. 2018; Georges et al. 2018). We set a call rate of 95%, which left 10,497 SNPs. A final 9732 SNPs were used in analyses after removing duplicated loci. In our PCoA, the two most informative principal coordinate axes described 8.9% and 4.3% of the variation between samples (Fig. 4). When plotted on these axes, the data reveal three clusters of graphed points. These three isolated genomic populations include: 1) Little Canoe Creek, 2) Ballplay Creek, and 3) Mill Creek, Coahulla Creek, and Coosawattee River. The Coosawattee River, Mill Creek, and Coahulla Creek populations comprised all individuals sampled from the upper Coosa River drainage in Georgia and Tennessee. The ancestry matrix returned $K = 3$ as the most supported number of ancestral lineages which are hereafter identified as MUs (cross entropy = 0.47; Fig. 2). The three MUs defined here match µsDNA data except that SNP data add samples from the Coosawattee River to the Conasauga and Coosawattee system MU. The population averages from the ancestry matrix were plotted on a map showing the geographic distribution of the three ancestral DNA types (Fig. 2). The Bayesian phylogeny resolved five monophyletic clades with no mixing of individuals from different localities (posterior probability > 0.95; Online Resource 1; Fig. S4).

Table 6 Effective population sizes (N_e) were estimated using microsatellite loci (µsDNA) and single nucleotide polymorphisms (SNPs) for *Etheostoma trisella* management units (MUs)

MU	SNPs		µsDNA	
	N_e [95% CI]	*N*	N_e [95% CI]	*N*
Little Canoe Creek	210 [208–212]	57	126 [51–∞]	46
Ballplay Creek	33 [32–33]	14	37 [17–243]	31
Conasauga and Coosawattee Rivers	103 [102–105]	22	84 [45–269]	54

[a]Confidence interval (CI), number of individuals (*N*)

The IBD function in R returned a significant positive association of genetic distance with geographic distance measured along waterways (Fig. 3; $r^2 = 0.62$; $p < 0.001$). Localities within the same creek systems were the most genetically similar (i.e., subpopulations within Little Canoe Creek and Ballplay Creek) followed by the Conasauga and Coosawattee creek systems, which are close together geographically. Ballplay Creek lies near the center of the known range of *E. trisella*, but the genetic distances between other creek systems are the highest among all comparisons (Fig. 3).

N_e estimates are displayed in Table 6, with Ballplay Creek being the smallest. The AMOVA computed in Arlequin (Table 2) showed that 11.53% of the total variation was among creek systems and 88.47% from individual variation within creek systems ($p < 0.05$); 10.83% of the total variation was among MUs (Table 3) and 89.17% from individual variation within MUs ($p < 0.05$). Additionally, pairwise population F_{st} values were calculated in Arlequin and plotted in a heat map with a dendrogram (Online Resource 1; Fig. S5;

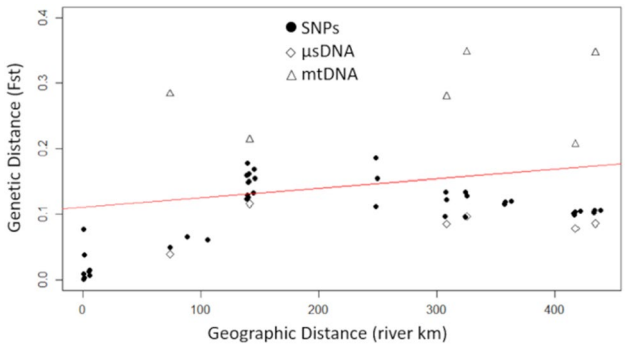

Fig. 3 Genetic distances (F_{st}) plotted against riverine distances (km) for *Etheostoma trisella* using mitochondrial DNA (mtDNA), microsatellite (µsDNA), and SNPs. IBD was calculated using mtDNA ($r^2 = 0.24$; $p = 0.29$), µsDNA ($r^2 = 0.34$; $p = 0.13$), and SNPs ($r^2 = 0.616$; p = 0.001). Results of IBD with significant support (i.e., SNPs) are plotted by the red line

Fig. 4 Principal Coordinate Analysis (PCoA) on 9732 single nucleotide polymorphisms (SNPs) from 93 *Etheostoma trisella* samples collected from five creek systems

all F_{st} $p < 0.05$). The F_{st} values were lowest between the Conasauga and Coosawattee creek systems (i.e., more gene flow) and highest between Little Canoe Creek and Ballplay Creek (i.e., less gene flow).

While there were no significant differences in H_o estimated from MUs (based on $K = 3$) versus creek systems, we report both sets of values (H_o µsDNA: $p = 0.91$; H_o SNPs: $p = 0.80$; Tables 4 and 5). Estimates of F_{st} were not significantly different based on µsDNA and SNPs ($p = 0.27$; Online Resource 1; Table S2; Fig. S5), but see below for disparities in F_{st} association with geographic distance. There were also no significant differences in N_e estimated with µsDNA and SNPs ($p = 0.61$; Table 6).

Discussion

Resources for the conservation of freshwater fishes are often distributed in response to the level of risk (USFWS 2018a). It is imperative to have guidance for allocating monetary or physical (fieldwork) resources. Using a multi-marker dataset, our µsDNA and SNP analyses converged to assign *E. trisella* populations to three distinct groups (MUs), with some occupied habitats disconnected by the construction of riverine impoundments. Further, we suggest that genetic isolation of these populations has occurred recently and likely as a result of habitat fragmentation from anthropogenic activities. The Ballplay Creek MU should be a priority for conservation management because N_e estimates (µsDNA = 37; SNPs = 33) are considerably lower, it has the lowest mean number of alleles per locus of all creek systems and MUs ($A = 4.71$), and shows signs of a recent bottleneck event.

We used the three data types to form a description of population structure for *E. trisella* in both recent and historical timelines (i.e., pre- and post-impoundment). While mitochondrial data are appropriate for detecting historical signatures of genetic changes (Anderson et al. 2010; Sanchez et al. 2020), µsDNA (Fluker et al. 2014a, b) and SNP data apply to contemporary population structure (Near et al. 2021). Our mtDNA data weakly support population structure for the species, show broad scale haplotype sharing, and did not return *E. trisella* populations as reciprocally monophyletic. We suggest that our mtDNA data provide evidence that populations of *E. trisella* were likely more broadly connected historically, or at least share a fairly recent history of gene flow. Our µsDNA and SNP data support low-moderate levels of structuring, indicating three distinct groups. The patterns of structure seen with genetic data have likely been driven by more contemporary events. It is expected that reciprocal monophyly of *E. trisella* populations would be reached by all data types, but that is not what we show here. In fact, this trend has been reported in other fish species when a multi-marker genetic approach was taken. Low levels of population structure were seen in populations of *Cyprinella gibbsi*, *E. tallapoosae*, and *E. tuscumbia* using mtDNA, but some populations were highly structured with µsDNA (Fluker et al. 2014a, b). Similarly, Near et al. (2021) saw resolution of population structure in *Percina kusha* and *P. freemanorum* using SNPs, but not with mtDNA markers.

Both µsDNA and SNP data showed no significant differences among creek systems or MUs for mean H_o nor allelic diversity. However, the Ballplay Creek population did show some evidence of a recent genetic bottleneck. Recent literature suggests that BOTTLENECK lacks power in detecting bottlenecks (Leblois et al. 2006; Girod et al. 2011), and it typically detects only population changes that are very recent or ongoing (Peery et al. 2012). Alternatively to this, the *M*-ratio method is thought to reveal older, more severe, or prolonged bottlenecks (Williamson-Natesan 2005; Girod et al. 2011); however, we did not find any support for bottlenecks with that method. The lower genetic diversity seen in µsDNA data from the Ballplay Creek population, although not significantly different from the other populations, may also be suggestive of a recent bottleneck, and significant losses in genetic diversity may not yet be realized.

Our data suggest that *E. trisella* populations show low-moderate population structure relative to comparable SNP and µsDNA datasets in the literature. Since estimates of heterozygosity are sensitive to the marker type, ascertainment method, and family structure, we note that µsDNA loci used here are amenable to interspecific comparison with data collected for related species that share homologous primer binding sites (Fluker et al. 2010, 2014a, 2019; McCall and Fluker 2021). Here we compare H_o means for *E. trisella* to other narrow, endemic darters and widespread generalists (Online Resource 1; Table S4). Because H_o estimates were significantly different between data types, we interpret them separately. The mean H_o estimated from µsDNA data for *E. trisella* creek systems and MUs ($H_o = 0.66$, Table 4; $H_o = 0.65$, Table 5) are very similar to that of other narrow endemics including *E. nuchale*, *E. pallididorsum*, *E. tuscumbia, E. cyanoprosopum,* and *E. boschungi* (mean H_o range: 0.55–0.70; Online Resource 1; Table S4; Fluker et al. 2010, 2014a, 2019, McCall and Fluker 2021). SNP loci for the narrow, endemic species *E. chienense* and *E.* cf. *oophylax* are comparable in quantity to our SNP dataset (Kattawar and Piller 2020). The mean SNP H_o for *E. trisella* creek systems and MUs ($H_o = 0.16$, Table 4; $H_o = 0.17$, Table 5) is higher than those for *E. chienense* ($H_o = 0.06$) and *E.* cf. *oophylax* ($H_o = 0.05$; Kattawar and Piller 2020). It is important to note that *E. chienense* is federally endangered, and it is expected that genetic diversity would be low for this species. Kattawar and Piller (2020) suggested that *E.* cf. *oophylax* showed H_o values similar to an endangered species because samples were collected from only a small portion of the range. *Etheostoma caeruleum, E. zonistium,* and *E. blennioides*

on the other hand are large-ranging species. *Etheostoma caeruleum* shows more genetic diversity than *E. trisella* in both SNP ($H_o = 0.32$; Oliveira et al. 2020) and µsDNA data ($H_o = 0.86$; Bossu and Near 2013). Similarly, populations of *E. blennioides* using SNPs ($H_o = 0.38$) and *E. zonistium* using µsDNA ($H_o = 0.80$) exhibited greater diversity than *E. trisella* (Malone et al. 2018; Fluker et al. 2019; see Online Resource 1; Table S4 for comparisons to more darter species). Considered in the context of previous studies, our results suggest that narrow endemic darter species, including *E. trisella* have lower genetic diversity than widespread generalist species. This is important because fishes with the same breeding behavior as *E. trisella* have limited gene pools, are prone to inbreeding, and are susceptible to further genetic diversity loss (McCall and Fluker 2021). Observed heterozygosity data did not coalesce to support a single MU or creek system as a target for management attention.

Comparable microsatellite data from the literature suggest that $F_{st} = 0.09$ (Table 2) for *E. trisella* indicates low-moderate levels of population structure. For example, Washburn et al. (2020) described low levels of genetic structure in the Tuxedo Darter (*E. lemniscatum;* $F_{st} = 0.01$); see also Heggenes and Røed (2006); Als et al. (2011); Feijoo et al. (2011); and Pujolar et al. (2014) for panmictic vertebrate populations with $F_{st} < 0.01$. Fluker et al. (2010, 2014a) showed moderate-high levels of population structure with $F_{st} > 0.20$ for three groundwater-dependent darters (see also Online Resource 1; Table S4).

Weiss Lake and H. Neely Henry Lake (Fig. 1) represent substantial physical barriers to gene flow among populations of small-bodied freshwater fishes. *E. trisella* localities in both Ballplay and Little Canoe creeks are in the Southern Shale Valley ecoregion (Fig. 1.), but Ballplay Creek has little topographic relief and is dominated by low-gradient streams (Johnson et al. 2013; O'Neil et al. 2022). The lower levels of genetic diversity in Ballplay Creek, in conjunction with evidence of a recent bottleneck, suggest that this population is likely suffering the effects of genetic drift in isolation. Alternatively, Ballplay Creek may have less suitable habitat for adults in the non-breeding system (low-gradient streams) compared to Little Canoe Creek (higher-gradient streams) and therefore hosts lower abundance and less genetic diversity when isolated by reservoirs. Coalescent simulations of six fish species have suggested that changes to genetic diversity caused by dams may be detectable in 40–60 generations (Ruzich et al. 2019) and as few as five generations in short-lived species (Murphy et al. 2008). Genetic diversity data collected from fish in streams impacted by impoundment also show effects in recent time (Skalski et al. 2008; Franssen 2011; Hudman and Gido 2013; Roberts et al. 2013; Fluker et al. 2014b). Simulation data suggest that anthropogenic habitat fragmentation can contribute to population isolation, and these recent effects may be confounded by riverine complexity (Brauer and Beheregaray 2020). While impoundment may not have a direct effect on population decline, Light and Marchetti (2007) suggest it may indirectly create altered habitats that support competitive invasive taxa. In addition, a study on Sockeye Salmon (*Oncorhynchus nerka*) suggests that while gene flow can decrease quickly following dam construction, life-history forms can also diverge over short time periods (Samarasin et al. 2017).

Our mtDNA and µsDNA data do not follow an IBD model (Wright 1943), but the SNP data show a significant positive association between genetic and geographic distance. Still, genetic distances were highest at intermediate geographic distances (i.e., comparisons with Ballplay Creek, $F_{st} = 0.10–0.19$). Although geographic distance may play a role in impeding gene flow, genetic diversity is highest at Ballplay Creek and the Coosawattee River at the center of the known distribution, not between the expected geographically distant creek systems. Fluker et al. (2014b) previously suggested that genetic structure of darters may be influenced to a greater degree by riverine impoundment despite the proximity of fish populations to one another (e.g., Little Canoe Creek and Ballplay Creek). These man made barriers are relatively recent constructions, and this lends more evidence to a recent isolation of *E. trisella* populations.

The USFWS (2017) has suggested that the primary threat to *E. trisella* populations is the loss of habitat that connects spawning and non-spawning areas. This is important because Brönmark et al. (2008) suggested that seasonal migration of freshwater fishes is an evolutionary adaptation to avoid predators and increase foraging success at various life-history stages. Similar to *E. trisella*, the Slackwater Darter (*E. boschungi*) exhibits brief annual breeding migrations into ephemeral habitats, and µsDNA data show evidence that breeding habitat may be an important factor in diversification of darters (Fluker et al. 2014a). Genetic data should be used in the future to investigate how behaviors like philopatry and dispersal affect genetic population structure in migratory fishes (Walton et al. 2021). While philopatry is not documented in *E. trisella*, site-specific migration of fishes with low genetic diversity to limited breeding habitats may cause further isolation. McCall and Fluker (2021) suggested that the survival of narrow, spring endemics is dependent on the persistence and connectivity of breeding and non-breeding habitat. If these species do migrate out of the local drainage, the chances of survival are low. When population numbers are low, spatial complexity is small, and the range is small, fish are not expected to withstand stochastic events (e.g., drought, landslides) or adapt to changing environmental conditions and catastrophic events (USFWS 2017). *E. trisella* meets multiple indicators for species that are expected to suffer future population size and diversity loss. We present evidence supporting three of the four MUs previously designated by the USFWS for *E. trisella* based on

genetic structuring. Additional assignment of the Coosawattee population to its own MU is supported because these fish are affected by hydrologic releases from the upstream dam not present in the Conasauga River system (Mill and Coahulla creeks). This spatial separation makes conservation strategies unique compared to the other MUs. Further, evidence of low N_e and A, plus a potential recent bottleneck event at Ballplay Creek emphasizes the need for attention to that MU. A better understanding of the distribution, abundance, and occupancy of all *E. trisella* MUs will be important for informing future decisions for the conservation of the species.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s10592-023-01572-9.

Acknowledgements The views and conclusions contained in this document are those of the authors and should not be interpreted as representing the opinions or policies of the U.S. Government, NFWF, or Southern Company. Mention of trade names or commercial products does not constitute their endorsement by the U.S. Government or NFWF or its funding sources. We thank Kenny Jones for preliminary analyses, Brett Albanese for help with grant management and donating tissues, and Peter J. Unmack for guidance with DArTseq™.

Author contributions The research project was designed by MWS, BLF, BRK, PEO, and SWM; samples were collected by BLF, BRK, and MSP; laboratory work and analyses were designed and conducted by KMF, BLF, MSP and MWS; and the paper was written by KMF with contributions from all the authors.

Funding This publication was made possible through a grant (62429) from the National Fish and Wildlife Foundation (NFWF), with funding provided by the U.S. Fish and Wildlife Service, the U.S. Forest Service, the University of Alabama, and Southern Company. A cost share was contributed by the Geological Survey of Alabama. This publication is a contribution of the Forest and Wildlife Research Center at Mississippi State University and is based upon work that is supported with McIntire-Stennis Funds, under a pending accession number.

Data availability Data associated with this study are available from the Dryad Digital Repository (https://doi.org/10.5061/dryad.jq2bvq8g6).

Declarations

Conflict of interest The authors have no conflicts of interest to declare.

References

Alcaraz C, Gholami Z (2020) Diversity and structure of fragmented populations of a threatened endemic cyprinodontid (*Aphanius sophiae*) inferred from genetics and otolith morphology: implications for conservation and management. J Zool Syst Evol Res 58:341–355. https://doi.org/10.1111/jzs.12333

Als TD, Hansen MM, Maes GE et al (2011) All roads lead to home: panmixia of European eel in the Sargasso Sea. Mol Ecol 20:1333–1346. https://doi.org/10.1111/j.1365-294X.2011.05011.x

Altshuler D, Pollara VJ, Cowles CR, Van Etten WJ, Baldwin J et al (2000) An SNP map of the human genome generated by reduced representation shotgun sequencing. Nature 407:513–516. https://doi.org/10.1038/35035083

Anderson C, Epperson B, Fortin M-J et al (2010) Considering spatial and temporal scale in landscape-genetic studies of gene flow. Mol Ecol 19:3565–3575. https://doi.org/10.1111/j.1365-294X.2010.04757.x

Bailey RM, Richards WJ (1963) Status of *Poecilichthys hopkinsi* Fowler and *Etheostoma trisella*, new species, percid fishes from Alabama, Georgia, and South Carolina. Occas Pap Mus Zool Univ Mich 630:1–21

Baird NA, Etter PD, Atwood TS, Currey MC, Shiver AL et al (2008) Rapid SNP discovery and genetic mapping using sequenced RAD markers. PLoS ONE 3:e3376. https://doi.org/10.1371/journal.pone.0003376

Bearden RA, McGregor SW, West DA et al (2021) A review of the distribution of the trispot darter (*Etheostoma trisella*) using environmental DNA and geospatial analysis. Geological Survey of Alabama Open-File Report 2105, Tuscaloosa, Alabama

Bohling J, Small M, Von Bargen J et al (2019) Comparing inferences derived from microsatellite and RADseq datasets: a case study involving threatened bull trout. Conserv Genet 20:329–342. https://doi.org/10.1007/s10592-018-1134-z

Boschung HT, Mayden RL (2004) Fishes of Alabama. Smithsonian Books, Washington, DC

Bossu CM, Near TJ (2013) Characterization of a contemporaneous hybrid zone between two darter species (*Etheostoma bison* and *E. caeruleum*) in the Buffalo River System. Genetica 141:75–88. https://doi.org/10.1007/s10709-013-9707-8

Brauer CJ, Beheregaray LB (2020) Recent and rapid anthropogenic habitat fragmentation increases extinction risk for freshwater biodiversity. Evol Appl 13:2857–2869. https://doi.org/10.1111/eva.13128

Brönmark C, Skov C, Brodersen J et al (2008) Seasonal migration determined by a trade-off between predator avoidance and growth. PLoS ONE 3:e1957. https://doi.org/10.1371/journal.pone.0001957

Clement M, Posada D, Crandall KA (2000) TCS: a computer program to estimate gene genealogies. Mol Ecol 9:1657–1659

Coleman RA, Gauffre B, Pavlova A et al (2018) Artificial barriers prevent genetic recovery of small isolated populations of a low-mobility freshwater fish. Heredity 120:515–532. https://doi.org/10.1038/s41437-017-0008-3

Corral-Lou A, Perea S, Doadrio I (2021) High genetic differentiation in the endemic and endangered freshwater fish *Achondrostoma salmantinum* Doadrio and Elvira, 2007 from Spain, as revealed by mitochondrial and SNP markers. Conserv Genet 22:585–600. https://doi.org/10.1007/s10592-021-01381-y

Do C, Waples RS, Peel D et al (2013) NeEstimator v2: re-implementation of software for the estimation of contemporary effective population size (N_e) from genetic data. Mol Ecol Resour 14:209–214. https://doi.org/10.1111/1755-0998.12157

Doadrio I, Perdices A, Machordom A (1996) Allozymic variation of the endangered killifish *Aphanius iberus* and its application to conservation. Environ Biol Fishes 45:259–271

Dray S, Dufour A-B (2007) The ade4 Package: implementing the duality diagram for ecologists. J Stat Softw 22:1–20. https://doi.org/10.18637/jss.v022.i04

Elshire RJ, Glaubitz JC, Sun Q, Poland JA, Kawamoto K et al (2011) A robust, simple genotyping-by-sequencing (GBS) approach for high diversity species. PLoS ONE 6(5):e19379. https://doi.org/10.1371/journal.pone.0019379

Endangered and Threatened Wildlife and Plants; Threatened Species Status for Trispot Darter, 83 FR 67131, 28 Dec 2018

Etnier DA, Starnes WC (1993) The Fishes of Tennessee. University of Tennessee Press, Knoxville

Evanno G, Regnaut S, Goudet J (2005) Detecting the number of clusters of individuals using the software structure: a simulation study. Mol Ecol 14:2611–2620. https://doi.org/10.1111/j.1365-294X.2005.02553.x

Excoffier L, Lischer HEL (2010) Arlequin (version 3.5.1.2): an integrated software package for population genetics data analysis. Evol Bioinform Online 1:47–50

Feijoo M, Lessa E, Loizaga de Castro R, Crespo E (2011) Mitochondrial and microsatellite assessment of population structure of South American sea lion (*Otaria flavescens*) in the southwestern Atlantic Ocean. Mar Biol 158:1857–1867. https://doi.org/10.1007/s00227-011-1697-4

Fischer J, Lindenmayer DB (2007) Landscape modification and habitat fragmentation: a synthesis. Glob Ecol Biogeogr 16:265–280. https://doi.org/10.1111/j.1466-8238.2007.00287.x

Fluker BL, Kuhajda BR, Lang NJ, Harris PM (2010) Low genetic diversity and small long-term population sizes in the spring endemic watercress darter, *Etheostoma nuchale*. Conserv Genet 11:2267–2279. https://doi.org/10.1007/s10592-010-0111-y

Fluker BL, Kuhajda BR, Harris PM (2014a) The influence of life-history strategy on genetic differentiation and lineage divergence in darters (Percidae: Etheostomatinae). Evolution 68:3199–3216. https://doi.org/10.1111/evo.12505

Fluker BL, Kuhajda BR, Harris PM (2014b) The effects of riverine impoundment on genetic structure and gene flow in two stream fishes in the Mobile River basin. Freshw Biol 59:526–543. https://doi.org/10.1111/fwb.12283

Fluker B, Jones K, Kuhajda B (2019) Genetic structure and diversity of the blueface darter *Etheostoma cyanoprosopum*, a microendemic freshwater fish in the southeastern USA. Endang Species Res 40:133–147. https://doi.org/10.3354/esr00986

Frankham R (2015) Genetic rescue of small inbred populations: meta-analysis reveals large and consistent benefits of gene flow. Mol Ecol 24:2610–2618. https://doi.org/10.1111/mec.13139

Franssen N (2011) Genetic structure of a native cyprinid in a reservoir-altered stream network. Freshw Biol 57:155–165. https://doi.org/10.1111/j.1365-2427.2011.02709.x

Fraser DJ, Bernatchez L (2001) Adaptive evolutionary conservation: towards a unified concept for defining conservation units. Mol Ecol 10:2741–2752. https://doi.org/10.1046/j.0962-1083.2001.01411.x

Freeman BJ, Hagler M (2016) Species profile for *Etheostoma trisella*, trispot darter. Georgia Biodiversity Portal. https://georgiabiodiversity.org/natels/profile?es_id=20635. Accessed 15 Feb 2023

Frichot E, François O (2015) LEA: an R package for landscape and ecological association studies. Methods Ecol Evol 6:925–929. https://doi.org/10.1111/2041-210X.12382

Gabel JM, Dakin EE, Freeman BJ, Porter BA (2008) Isolation and identification of eight microsatellite loci in the Cherokee darter (*Etheostoma scotti*) and their variability in other members of the genera *Etheostoma*, *Ammocrypta*, and *Percina*. Mol Ecol Resour 8:149–151. https://doi.org/10.1111/j.1471-8286.2007.01903.x

Garza JC, Williamson EG (2001) Detection of reduction in population size using data from microsatellite loci. Mol Ecol 10:305–318. https://doi.org/10.1046/j.1365-294X.2001.01190.x

Georges A, Gruber B, Pauly GB et al (2018) Genomewide SNP markers breathe new life into phylogeography and species delimitation for the problematic short-necked turtles (Chelidae: *Emydura*) of eastern Australia. Mol Ecol 27:5195–5213. https://doi.org/10.1111/mec.14925

Georgia Department of Natural Resources (2023) Rare species status maps, *Etheostoma trisella* (trispot darter). Georgia Biodiversity Portal. https://georgiabiodiversity.org/natels/rangemaps?es_id=20635. Accessed 15 Feb 2023

Girod C, Vitalis R, Leblois R, Fréville H (2011) Inferring population decline and expansion from microsatellite data: a simulation-based evaluation of the Msvar method. Genetics 188:165–179. https://doi.org/10.1534/genetics.110.121764

Gruber B, Unmack PJ, Berry OF, Georges A (2018) dartR: an r package to facilitate analysis of SNP data generated from reduced representation genome sequencing. Mol Ecol Resour 18:691–699. https://doi.org/10.1111/1755-0998.12745

Haddad NM, Brudvig LA, Clobert J et al (2015) Habitat fragmentation and its lasting impact on Earth's ecosystems. Sci Adv 1:e1500052. https://doi.org/10.1126/sciadv.1500052

Hall T (1999) BioEdit: a user-friendly biological sequence alignment editor and analysis program for Windows 95/98/NT. Nucleic Acids Symp Ser 41:95–98

Hall T, Alzohairy DAM (2011) BioEdit: an important software for molecular biology. GERF Bull Biosci 2:60–61

Heggenes J, Røed KH (2006) Do dams increase genetic diversity in brown trout (*Salmo trutta*)? Microgeographic differentiation in a fragmented river. Ecol Freshw Fish 15:366–375. https://doi.org/10.1111/j.1600-0633.2006.00146.x

Hudman S, Gido K (2013) Multi-scale effects of impoundments on genetic structure of creek chub (*Semotilus atromaculatus*) in the Kansas River basin. Freshw Biol 58:441–453. https://doi.org/10.1111/fwb.12079

Jelks HL, Walsh SJ, Burkhead NM et al (2008) Conservation status of imperiled North American freshwater and diadromous fishes. Fisheries 33:372–407. https://doi.org/10.1577/1548-8446-33.8.372

Johansen T, Besnier F, Quintela M et al (2020) Genomic analysis reveals neutral and adaptive patterns that challenge the current management regime for East Atlantic cod *Gadus morhua* L. Evol Appl 13:2673–2688. https://doi.org/10.1111/eva.13070

Johnson CC, O'Neil PE, Shepard TE et al (2013) Status survey of the trispot darter, *Etheostoma trisella*, in Alabama 2008–2012. Geological Survey of Alabama Open-File Report 1302, Tuscaloosa, AL

Kalinowski ST (2005) hp-rare 1.0: a computer program for performing rarefaction on measures of allelic richness. Mol Ecol Notes 5:187–189. https://doi.org/10.1111/j.1471-8286.2004.00845.x

Kattawar J, Piller KR (2020) Comparative population genetics of the federally endangered relict darter, and its sister taxon the clarks darter (Teleostei: Percidae). Conserv Genet 21:957–970. https://doi.org/10.1007/s10592-020-01300-7

Kilian A, Wenzl P, Huttner E et al (2012) Diversity arrays technology: a generic genome profiling technology on open platforms. In: Pompanon F, Bonin A (eds) Data production and analysis in population genomics: methods and protocols. Humana Press, Totowa, pp 67–89

Kim KS, Roe KJ (2021) Genome-wide SNPs redefines species boundaries and conservation units in the freshwater mussel genus *Cyprogenia* of North America. Sci Rep 11:10752. https://doi.org/10.1038/s41598-021-90325-0

Knaepkens G, Bervoets L, Verheyen E, Eens M (2004) Relationship between population size and genetic diversity in endangered populations of the European bullhead (*Cottus gobio*): implications for conservation. Biol Conserv 115:403–410. https://doi.org/10.1016/S0006-3207(03)00156-3

Lang NJ, Mayden RL (2007) Systematics of the subgenus *Oligocephalus* (Teleostei: Percidae: *Etheostoma*) with complete subgeneric sampling of the genus *Etheostoma*. Mol Phylogenet Evol 43:605–615. https://doi.org/10.1016/j.ympev.2006.09.022

Leblois R, Estoup A, Streiff R (2006) Genetics of recent habitat contraction and reduction in population size: does isolation by distance matter? Habitat reduction of IBD populations. Mol Ecol 15:3601–3615. https://doi.org/10.1111/j.1365-294X.2006.03046.x

Light T, Marchetti MP (2007) Distinguishing between invasions and habitat changes as drivers of diversity loss among California's

freshwater fishes. Conserv Biol 21:434–446. https://doi.org/10.1111/j.1523-1739.2006.00643.x

Lopes-Lima M, Sousa R, Geist J et al (2017) Conservation status of freshwater mussels in Europe: state of the art and future challenges. Biol Rev 92:572–607. https://doi.org/10.1111/brv.12244

Magris RA, Costa MDP, Ferreira CEL et al (2021) A blueprint for securing Brazil's marine biodiversity and supporting the achievement of global conservation goals. Divers Distrib 27:198–215. https://doi.org/10.1111/ddi.13183

Malone EW, Perkin JS, Leckie BM et al (2018) Which species, how many, and from where: Integrating habitat suitability, population genomics, and abundance estimates into species reintroduction planning. Glob Change Biol 24:3729–3748. https://doi.org/10.1111/gcb.14126

Mandimbihasina AR, Frasier CL, Hagenson RA et al (2020) Conservation genetics of Madagascar's critically endangered ploughshare tortoise (*Astrochelys yniphora*). Conserv Genet 21:109–121. https://doi.org/10.1007/s10592-019-01236-7

McCall BL, Fluker BL (2021) Conservation genetic assessment of the paleback darter, *Etheostoma pallididorsum*, a narrowly distributed endemic in the Ouachita Highlands, Arkansas, USA. Aquat Conserv 31:1817–1830. https://doi.org/10.1002/aqc.3510

Mettee MF, O'Neil PE, Pierson JM (1996) Fishes of Alabama and the Mobile Basin. Oxmoor House, Birmingham

Murphy M, Evans J, Cushman S, Storfer A (2008) Representing genetic variation as continuous surfaces: an approach for identifying spatial dependency in landscape genetic studies. Ecography 31:685–697. https://doi.org/10.1111/j.1600-0587.2008.05428.x

Near TJ, Bossu CM, Bradburd GS et al (2011) Phylogeny and temporal diversification of darters (Percidae: Etheostomatinae). Syst Biol 60:565–595. https://doi.org/10.1093/sysbio/syr052

Near TJ, MacGuigan DJ, Boring EL et al (2021) A new species of bridled darter endemic to the Etowah River system in Georgia (Percidae: Etheostomatinae: *Percina*). Bull Peabody Mus Nat Hist 62:15–42. https://doi.org/10.3374/014.062.0102

Neely DA, Mayden RL (1999) Status survey of the trispot darter (*Etheostoma trisella*) in Alabama. Alabama Department of Conservation and Natural Resources, Montgomery

O'Neil PE, Shepard TE, Johnson CC et al (2009) Rediscovery of the trispot darter, *Etheostoma trisella*, in Alabama. Geological Survey of Alabama open-file report 0909, Tuscaloosa, AL

O'Neil PE, Wynn AE, Pierce GH (2022) A review of the life history, distribution, analysis of current threats, and discussion of conservation actions for the trispot darter (*Etheostoma trisella*). Geological Survey of Alabama Bulletin 191, Tuscaloosa, AL

Oliveira DR, Reid BN, Fitzpatrick SW (2020) Genome-wide diversity and habitat underlie fine-scale phenotypic differentiation in the rainbow darter (*Etheostoma caeruleum*). Evol Appl 14:498–512. https://doi.org/10.1111/eva.13135

Page LM, Burr BM (2011) Peterson field guide to freshwater fishes of North America north of Mexico. Houghton Mifflin Harcourt, Boston

Peery MZ, Kirby R, Reid BN et al (2012) Reliability of genetic bottleneck tests for detecting recent population declines. Mol Ecol 21:3403–3418. https://doi.org/10.1111/j.1365-294X.2012.05635.x

Pereira HM, Leadley PW, Proença V et al (2010) Scenarios for global biodiversity in the 21st century. Science 330:1496–1501. https://doi.org/10.1126/science.1196624

Piry S, Luikart G, Cornuet JM (1999) BOTTLENECK: a computer program for detecting recent reductions in the effective population size using allele frequency data. J Hered 90:502–503. https://doi.org/10.1093/jhered/90.4.502

Pritchard JK, Stephens M, Donnelly P (2000) Inference of population structure using multilocus genotype data. Genetics 155:945–959. https://doi.org/10.1093/genetics/155.2.945

Pujolar JM, Jacobsen MW, Als TD et al (2014) Genome-wide single-generation signatures of local selection in the panmictic European eel. Mol Ecol 23:2514–2528. https://doi.org/10.1111/mec.12753

R Core Team (2021) R: a language and environment for statistical computing. R Foundation for Statistical Computing, Vienna

Rambaut A, Drummond AJ, Xie D et al (2018) Posterior summarization in Bayesian phylogenetics using Tracer 1.7. Syst Biol 67:901–904. https://doi.org/10.1093/sysbio/syy032

Rands M, Adams W, Bennun L et al (2010) Biodiversity conservation: challenges beyond 2010. Science 329:1298–1303. https://doi.org/10.1126/science.1189138

Roberts J, Angermeier P, Hallerman E (2013) Distance, dams and drift: what structures populations of an endangered, benthic stream fish? Freshw Biol 58:2050–2064. https://doi.org/10.1111/fwb.12190

Ruzich J, Turnquist K, Nye N et al (2019) Isolation by a hydroelectric dam induces minimal impacts on genetic diversity and population structure in six fish species. Conserv Genet 20:1421–1436. https://doi.org/10.1007/s10592-019-01220-1

Ryon MG (1986) The life history and ecology of *Etheostoma trisella* (Pisces: Percidae). Am Midl Nat 115:73–86. https://doi.org/10.2307/2425838

Samarasin P, Shuter BJ, Rodd FH (2017) After 100 years: hydroelectric dam-induced life-history divergence and population genetic changes in sockeye salmon (*Oncorhynchus nerka*). Conserv Genet 18:1449–1462. https://doi.org/10.1007/s10592-017-0992-0

Sanchez G, Kawai K, Yamashiro C et al (2020) Patterns of mitochondrial and microsatellite DNA markers describe historical and contemporary dynamics of the humboldt squid *Dosidicus gigas* in the Eastern Pacific Ocean. Rev Fish Biol Fish 30:519–533. https://doi.org/10.1007/s11160-020-09609-9

Shelton-Nix E (2017) Alabama wildlife. The University of Alabama Press, Tuscaloosa

Skalski G, Landis J, Grose M, Hudman S (2008) Genetic structure of creek chub, a headwater minnow, in an impounded river system. Trans Am Fish Soc 137:962–975. https://doi.org/10.1577/T07-060.1

Suchard MA, Lemey P, Baele G et al (2018) Bayesian phylogenetic and phylodynamic data integration using BEAST 1.10. Virus Evol 4:vey016. https://doi.org/10.1093/ve/vey016

Switzer JF, Welsh SA, King TL (2008) Microsatellite DNA primers for the candy darter, *Etheostoma osburni* and variegate darter, *Etheostoma variatum*, and cross-species amplification in other darters (Percidae). Mol Ecol Resour 8:335–338. https://doi.org/10.1111/j.1471-8286.2007.01946.x

Tonnis BD (2006) Microsatellite DNA markers for the rainbow darter, *Etheostoma caeruleum* (Percidae), and their potential utility for other darter species. Mol Ecol Notes 6:230–232. https://doi.org/10.1111/j.1471-8286.2005.01203.x

Tyers M (2022) riverdist: river network distance computation and applications. https://CRAN.R-project.org/package=riverdist. Accessed 15 Feb 2023

U.S. Fish and Wildlife Service (2017) Species status assessment for the trispot darter (*Etheostoma trisella*). U.S. Fish and Wildlife Service, Atlanta

U.S. Fish and Wildlife Service (2018) Endangered and threatened wildlife and plants; threatened species status for trispot darter. U.S. Fish and Wildlife Service, Atlanta

U.S. Fish and Wildlife Service (2018) Recovery outline for trispot darter (*Etheostoma trisella*). U.S. Fish and Wildlife Service, Atlanta

U.S. Fish and Wildlife Service (2020) Endangered and threatened wildlife and plants; designation of critical habitat for the trispot darter. US Fish and Wildlife Service, Atlanta

Walton Z, Hagenlund M, Østbye K (2021) Moving far, staying close: red fox dispersal patterns revealed by SNP genotyping. Conserv Genet 22:249–257. https://doi.org/10.1007/s10592-021-01332-7

Warnes G, Bolker B, Bonebakker L et al (2016) gplots: various r programming tools for plotting data. http://CRAN.R-project.org/package=gplots. Accessed 15 Feb 2023

Warren ML (2004) Etheostoma trisella. In: Mirarchi RE (ed) Alabama wildlife: imperiled aquatic mollusks and fishes. The University of Alabama Press, Tuscaloosa, p 173

Washburn BA, Cashner MF, Blanton RE (2020) Small fish, large river: surprisingly minimal genetic structure in a dispersal-limited, habitat specialist fish. Ecol Evol 10:2253–2268. https://doi.org/10.1002/ece3.6064

Willi Y, van Buskirk J, Hoffmann AA (2006) Limits to the adaptive potential of small populations. Annu Rev Ecol Evol Syst 37:433–458. https://doi.org/10.1146/annurev.ecolsys.37.091305.110145

Williams J, Robison H (1980) *Ozarka*, a new subgenus of *Etheostoma* (Pisces: Percidae). Brimleyana 4:149–156

Williamson-Natesan EG (2005) Comparison of methods for detecting bottlenecks from microsatellite loci. Conserv Genet 6:551–562. https://doi.org/10.1007/s10592-005-9009-5

Withers DI (2016) A guide to the rare animals of Tennessee. Division of Natural Areas, Tennessee Department of Environment & Conservation, Nashville

Wright S (1943) Isolation by distance. Genetics 28:114–138

Wu H, Chen J, Xu J et al (2019) Effects of dam construction on biodiversity: a review. J Clean Prod 221:480–489. https://doi.org/10.1016/j.jclepro.2019.03.001

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.